Indexing and query processing

Data engineering · University · Lecture 2



In this lecture

- Explain B-tree indexes.
- Reason about selectivity.
- Read query plans.

Remember

- Wide trees stay shallow.
- Writes pay for reads.
- EXPLAIN before and after.

Tune one query

1. Warm-up (5 min): Guess the cost
2. Explain (15 min): The slow plan.
3. Index (15 min): Add and compare.
4. Measure (20 min): Writes.
5. Wrap-up (5 min): One-minute paper.

Let's discuss

1. How many pages did the index read?
2. When is a scan cheaper?
3. What did writes cost?

Exit ticket

- Name one column in your system you would index and why.